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NEXUS - Next-Gen AI Agent

A powerful AI-powered desktop automation and RPA (Robotic Process Automation) agent that understands natural language commands to control your computer, manage files, manipulate Excel spreadsheets, and more.

Features

- 🗣️ **Natural Language Commands** - Just type or speak what you want to do
- 📁 **File Operations** - Open, move, rename, delete files with voice/text commands
- 📊 **Excel Manipulation** - Read, write, add rows, apply styles to spreadsheets
- 🖥️ **App Control** - Launch and close any application
- 🔍 **Smart File Search** - Fast cached search across your system
- 📄 **Document Processing** - Extract text from PDF/Word, convert formats

- 🌐 **AI Web Automation** - Agent can browse, search, and extract data from the web using Playwright

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🚀 Quick Start

Prerequisites

Requirement	Version
Python	3.10+
Node.js	18+
npm	9+

Step 1: Clone the Repository

```
git clone https://github.com/Rajkumars777/AI-agent---LTID.git
cd AI-agent---LTID
```

Step 2: Setup Backend

```
# Navigate to src
cd src

# Create virtual environment (recommended)
python -m venv venv
.\venv\Scripts\activate  # Windows
# source venv/bin/activate  # Linux/Mac

# Install Python dependencies
pip install -r requirements.txt

# Install Playwright browsers (for web automation)
playwright install
```

Step 3: Configure Environment

Create a `.env` file in the `src/` folder:

```
# Required: Choose ONE LLM provider

# Option 1: OpenRouter (Recommended - works with many models)
OPENROUTER_API_KEY=your_openrouter_api_key

# Option 2: Google Gemini
GEMINI_API_KEY=your_gemini_api_key
```

Get your API key:

- **OpenRouter**: https://openrouter.ai/keys (supports GPT-4, Claude, Llama, etc.)
- **Gemini**: https://makersuite.google.com/app/apikey

Step 4: Run Backend

```
cd src
python main.py
```

✅ Backend runs on: `http://localhost:8000`

Step 5: Setup & Run Frontend

Open a **new terminal**:

```
cd frontend

# Install Node dependencies
npm install

# Start development server
npm run dev
```

✅ Frontend runs on: `http://localhost:3000`

Step 6: Use the App

1. Open `http://localhost:3000` in your browser
2. Type or click the mic to speak commands like:
    - `"open notepad"`
    - `"delete sample.xlsx"`
    - `"rename old.txt to new.txt"`
    - `"add a row to budget.xlsx with name John and amount 500"`

📦 Dependencies

Backend (Python)

Package	Purpose
`fastapi`	Web framework for API
`uvicorn`	ASGI server
`python-dotenv`	Environment variables
`pyautogui`	Desktop automation
`pygetwindow`	Window management
`openpyxl`	Excel read/write
`pandas`	Data manipulation
`polars`	Fast data processing
`pymupdf`	PDF text extraction
`python-docx`	Word document handling
`playwright`	Browser automation
`faster-whisper`	Local voice-to-text (free)
`send2trash`	Safe file deletion
`AppOpener`	Application launching
`langchain`	LLM framework
`langgraph`	Agent workflows

Frontend (Node.js)

Package	Purpose
`next`	React framework
`react`	UI library
`tailwindcss`	Styling
`framer-motion`	Animations
`lucide-react`	Icons
`axios`	API calls

📂 Project Structure

```
AI-agent---LTID/
├── src/                        # Python FastAPI Backend
│   ├── main.py                 # 🚀 Entry point - FastAPI app
│   ├── agent.py                # 🧠 Main agent logic & command routing
│   ├── requirements.txt        # Python dependencies
│   ├── .env                    # API keys (create this)
│   │
│   ├── capabilities/           # 🔧 Feature modules
│   │   ├── desktop.py          # File ops, app control, file search
│   │   ├── excel_manipulation.py # Excel read/write/style
│   │   ├── document.py         # PDF/Word extraction + conversions
│   │   ├── browser.py          # Web automation (Playwright)
│   │   ├── data.py             # Data processing (Polars)
│   │   ├── code_generator.py   # Dynamic code generation
│   │   ├── dictation.py        # Voice-to-text (Faster-Whisper)
│   │   ├── file_search_cache.py # Fast file search with caching
│   │   └── llm_general.py      # LLM utilities
│   │
│   ├── execution/              # 🔄 Workflow & NLU
│   │   ├── nlu.py              # Natural Language Understanding
│   │   ├── gemini_adapter.py   # Google Gemini LLM
│   │   ├── openrouter_adapter.py # OpenRouter multi-model
│   │   └── system_utils.py     # System utilities
│   │
│   ├── routers/                # 🌐 API Routes
│   │   ├── agent.py            # /agent/* endpoints
│   │   └── tools.py            # /tools/* endpoints
│   │
│   ├── api/                    # Voice API
│   │   └── dictation.py        # Voice transcription endpoint
│   │
```

```
│   └── tests/                  # 🧪 Test files
│       └── test_*.py
│
├── frontend/                   # Next.js React Frontend
│   ├── src/
│   │   ├── app/                # Next.js pages
│   │   │   └── page.tsx        # Main page
│   │   ├── components/         # React components
│   │   │   ├── InputConsole.tsx # Command input + mic
│   │   │   ├── TimelineFeed.tsx # Command history
│   │   │   └── ...
│   │   └── styles/             # CSS styles
│   ├── package.json
│   └── tailwind.config.js
│
└── README.md                   # This file
```

💡 Example Commands

Command	What it does
`open notepad`	Launches Notepad
`open excel`	Launches Microsoft Excel
`close chrome`	Closes Chrome browser
`delete report.pdf`	Moves file to Recycle Bin
`rename old.txt to new.txt`	Renames the file
`move data.xlsx to Documents`	Moves file to Documents folder
`read sample.xlsx`	Shows Excel contents
`add row to data.xlsx with name John and age 25`	Adds a new row

🛠️ Troubleshooting

Backend won't start

```
# Make sure virtual environment is activated
.\venv\Scripts\activate

# Reinstall dependencies
pip install -r requirements.txt
```

Frontend errors

```
# Clear npm cache and reinstall
rm -rf node_modules
npm install
```

"API key not found" error

- Make sure `.env` file exists in `src/` folder
- Check that API key is correctly set (no extra spaces)

🤝 Contributing

1. Fork the repository
2. Create a feature branch
3. Make your changes
4. Submit a pull request

Made with ❤️ by Rajkumar